

Exercise-3: Stored Procedures

exercise3.sql

-- Accounts Table

```
CREATE TABLE Accounts (  
    AccountID NUMBER PRIMARY KEY,  
    CustomerName VARCHAR2(100),  
    AccountType VARCHAR2(20),  
    Balance NUMBER(10,2)  
);
```

-- Employees Table

```
CREATE TABLE Employees (  
    EmployeeID NUMBER PRIMARY KEY,  
    EmployeeName VARCHAR2(100),  
    Department VARCHAR2(50),  
    Salary NUMBER(10,2)  
);
```

-- Inserting Data

```
INSERT INTO Accounts VALUES (101,'Aditya','Savings',10000);  
INSERT INTO Accounts VALUES (102,'Rahul','Savings',15000);  
INSERT INTO Accounts VALUES (103,'Priya','Current',20000);  
INSERT INTO Accounts VALUES (104,'Neha','Savings',5000);
```

```
INSERT INTO Employees VALUES (1,'Amit','IT',50000);
```

```
INSERT INTO Employees VALUES (2,'Sneha','IT',60000);
```

```
INSERT INTO Employees VALUES (3,'Raj','HR',45000);
```

```
COMMIT;
```

```
-- Scenario 1:
```

```
CREATE OR REPLACE PROCEDURE ProcessMonthlyInterest
```

```
AS
```

```
BEGIN
```

```
    UPDATE Accounts
```

```
    SET Balance = Balance * 1.01
```

```
    WHERE AccountType = 'Savings';
```

```
COMMIT;
```

```
END;
```

```
/
```

```
-- Execution
```

```
EXEC ProcessMonthlyInterest;
```

```
-- Verification
```

```
SELECT * FROM Accounts;
```

Output (Scenario 1):

```
Table ACCOUNTS created.
```

```
Table EMPLOYEES created.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
1 row inserted.
```

```
Commit complete.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE QUERY RESULT

All rows fetched: 4 in 0.841 seconds

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE	
1	101	Aditya	Savings	10100	
2	102	Rahul	Savings	15150	
3	103	Priya	Current	20000	
4	104	Neha	Savings	5050	

Continued – Scenario 2:

-- Scenario 2:

```
CREATE OR REPLACE PROCEDURE UpdateEmployeeBonus(
```

```
    p_department IN VARCHAR2,
```

```
    p_bonus_percent IN NUMBER
```

```
)
```

```
AS
```

```
BEGIN
```

```
    UPDATE Employees
```

```
    SET Salary = Salary +
```

```
        (Salary * p_bonus_percent / 100)
```

```
    WHERE Department = p_department;
```

```
    COMMIT;
```

```
END;
```

```
/
```

-- Execution

```
EXEC UpdateEmployeeBonus('IT',10);
```

-- Verification

```
SELECT * FROM Employees;
```

Output (Scenario 2):

```
Procedure PROCESSMONTHLYINTEREST compiled
```

```
PL/SQL procedure successfully completed.
```

```
Procedure UPDATEEMPLOYEEBONUS compiled
```

```
PL/SQL procedure successfully completed.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE

All rows fetched: 3 in 0.022 seconds

	EMPLOYEEID	EMPLOYEENAME	DEPARTMENT	SALARY
1	1	Amit	IT	55000
2	2	Sneha	IT	66000
3	3	Raj	HR	45000

Continued - Scenario 3:

-- Scenario 3:

```
CREATE OR REPLACE PROCEDURE TransferFunds(
```

```
    p_from_account IN NUMBER,
```

```
    p_to_account   IN NUMBER,
```

```
    p_amount       IN NUMBER
```

```
)
```

```
AS
```

```
    v_balance NUMBER;
```

```
BEGIN
```

```
    SELECT Balance
```

INTO v_balance

FROM Accounts

WHERE AccountID = p_from_account;

IF v_balance >= p_amount THEN

UPDATE Accounts

SET Balance = Balance - p_amount

WHERE AccountID = p_from_account;

UPDATE Accounts

SET Balance = Balance + p_amount

WHERE AccountID = p_to_account;

COMMIT;

DBMS_OUTPUT.PUT_LINE(

 'Transfer Successful'

);

ELSE

DBMS_OUTPUT.PUT_LINE(

 'Insufficient Balance'

);

```
END IF;
```

```
END;
```

```
/
```

```
-- Output
```

```
SET SERVEROUTPUT ON;
```

```
-- Execution
```

```
EXEC TransferFunds(101,102,2000);
```

```
-- Verification
```

```
SELECT * FROM Accounts;
```

Output (Scenario 3):

```
Procedure TRANSFERFUNDS compiled

Transfer Successful

PL/SQL procedure successfully completed.
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS GITLENS SONARQUBE QUERY RESULT

All rows fetched: 4 in 0.018 seconds

	ACCOUNTID	CUSTOMERNAME	ACCOUNTTYPE	BALANCE	
1	101	Aditya	Savings	8100	
2	102	Rahul	Savings	17150	
3	103	Priya	Current	20000	
4	104	Neha	Savings	5050	